Practical-4	*Implementation and Time analysis of factorial program using iterative and recursive method*
Date :	

1)Iterative

```
FACTORIAL-ITERATIVE
    fact ← 1
    FOR i ← 1 TO n DO
        fact ← fact * i
    END FOR
    RETURN fact
END FACTORIAL-ITERATIVE
```

code:

```
#include <iostream>
using namespace std;

long long factorialIterative(int n) {
    long long fact = 1;

    for (int i = 1; i <= n; i++) {
        fact = fact * i;
    }

    return fact;
}

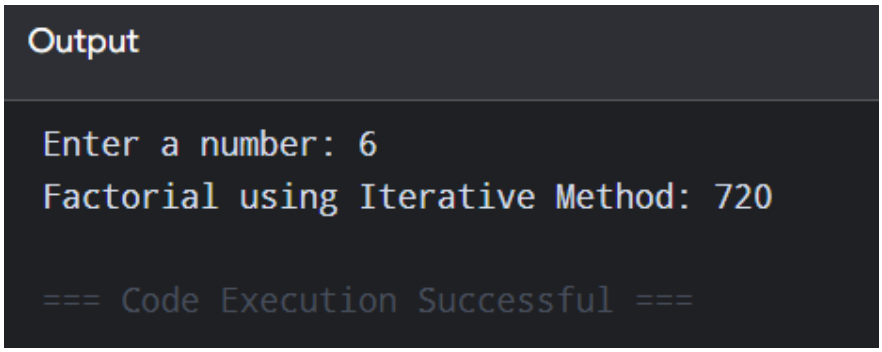
int main() {
    int n;

    cout << "Enter a number: ";
    cin >> n;

    cout << "Factorial using Iterative Method: "
         << factorialIterative(n);

    return 0;
}
```

Output:



Time complexity:

iterative :-

$T(n) = 1 + (n+1) + n + n$

$T(n) = 3n + 2$

$T(n) = O(n)$

Best case : $O(n)$

Avg case : $O(n)$

Worst case : $O(n)$

2)Recursion

```
FACTORIAL-RECURSIVE(n)
    IF n=0 OR n=1 THEN
        RETURN 1
    END IF
    RETURN n*FACTORIAL-RECURSIVE(n-1)
END FACTORIAL-RECURSIVE
```

code:

```
#include <iostream>
using namespace std;

long long factorialRecursive(int n) {
    if (n == 0 || n == 1)
        return 1;

    return n * factorialRecursive(n - 1);
}

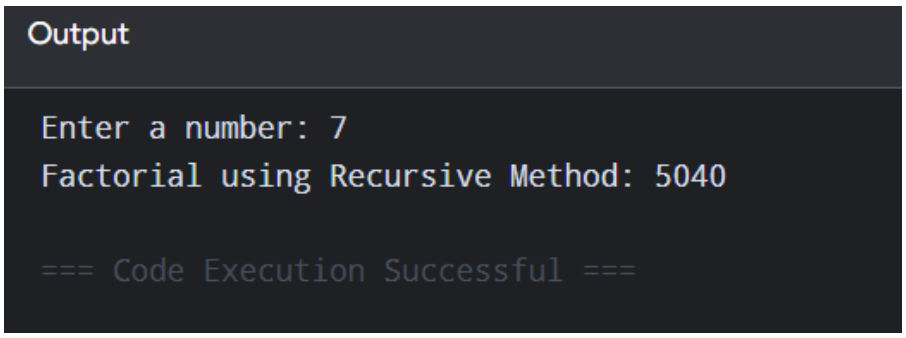
int main() {
    int n;

    cout << "Enter a number: ";
    cin >> n;

    cout << "Factorial using Recursive Method: "
         << factorialRecursive(n);

    return 0;
}
```

output:



Time complexity:

Recursive :-

$T(n) = T(n-1) + c$

$= T(n-2) + 2c$

$= T(n-3) + 3c$

$T(n) = T(n-1) + nc$

$T(n) = O(n)$

Best case : $O(n)$

Avg case : $O(n)$

Worst case : $O(n)$

Conclusion:

The factorial of a given number was successfully implemented using both iterative and recursive methods. The iterative method uses a loop to calculate the factorial, while the recursive method calculates it by repeatedly calling the same function until the base condition is reached. Both methods have a time complexity of O(n). However, the iterative method requires O(1) space, whereas the recursive method requires O(n) space due to the recursive function call stack. Thus, the iterative method is more memory-efficient, while the recursive method provides a simpler and more intuitive implementation.